

USER INFORMATION NEBULISER SOLUTION

SALBUNEB (Salbutamol 2.5 mg/ 2.5 mL Nebuliser Solution)

Composition:

Each 2.5 mL SALBUNEB contains:

Salbutamol Sulphate 3.0 mg equivalent to Salbutamol 2.5 mg in normal saline solutionq.s.

Characteristic:

SALBUNEB is a sterile, clear and colourless solution.

Pharmacodynamics:

Salbutamol is a selective Beta₂-agonist providing shortacting (4-6 hour) bronchodilation with a fast onset (within 5 minutes) in reversible airways obstruction. At therapeutic doses it acts on the Beta₂-adrenoceptors of bronchial muscle. With its fast onset of action, it is particularly suitable for the management and prevention of attack in asthma.

Pharmacokinetics:

Salbutamol administered intravenously has a half-life of 4 to 6 hours and is cleared partly renally, and partly by metabolism to the inactive 4'-O-sulphate (phenolic sulphate) which is also excreted primarily in the urine. The faeces are a minor route of excretion. Most of a dose of salbutamol given intravenously, orally or by inhalation is excreted within 72 hours. Salbutamol is bound to plasma proteins to the extent of 10%.

After administration by the inhaled route between 10 and 20% of the dose reaches the lower airways. The remainder is retained in the delivery system or is deposited in the oropharynx from where it is swallowed.

The fraction deposited in the airways is absorbed into the pulmonary tissues and circulation, but is not metabolised by the lung. On reaching the systemic circulation it becomes accessible to hepatic metabolism and is excreted, primarily in the urine, as unchanged drug and as the phenolic sulphate.

The swallowed portion of an inhaled dose is absorbed from the gastrointestinal tract and undergoes considerable first-pass metabolism to the phenolic sulphate. Both unchanged drug and conjugate are excreted primarily in the urine.

Indications:

SALBUNEB is indicated for use in the routine management of chronic bronchospasm unresponsive to conventional therapy, and in the treatment of acute severe asthma.

Recommended Dosage:

Adults (including the elderly): 2.5 mg to 5 mg Salbutamol up to four times a day. Up to 40 mg per day can be given under strict medical supervision in hospital.

Children 4 years and above: 2.5 mg to 5 mg up to four times a day.

Children 18 months to 4 years: 2.5 mg up to four times a day. The dose may be increased to 5 mg if necessary, but medical assessment should be considered since alternative therapy may be indicated.

Infants under 18 months old: 1.25 mg (0.25 mg/kg) to 2.5 mg up to four times a day. As transient hypoxia may occur supplemental oxygen therapy should be considered.

SALBUNEB is intended to be used undiluted. However, if prolonged delivery time (more than 10 minutes) is required, the solution may be diluted with sterile normal saline.

Route of Administration:

Unit Dose Vial: The unit dose vials are intended only for inhalation with suitable nebulising devices and should not be taken orally or administered parenterally.

Contraindications:

Although intravenous Salbutamol and, occasionally, Salbutamol tablets are used in the management of premature labour uncomplicated by conditions such as placenta praevia, antepartum haemorrhage, or toxæmia of pregnancy, inhaled Salbutamol preparations are not appropriate for managing premature labour. Salbutamol preparations should not be used for threatened abortion. SALBUNEB is contraindicated in patients with a history of hypersensitivity to any of the components.

Warnings and Precautions:

SALBUNEB must only be used by inhalation, to be breathed in through the mouth, and must not be injected or swallowed.

Bronchodilators should not be the only or main treatment in patients with severe or unstable asthma. Severe asthma requires regular medical assessment, including lung function testing, as patients are at risk of severe attacks and even death. Physicians should consider using the maximum recommended dose of inhaled corticosteroid and/or oral corticosteroid therapy in these patients.

Patients receiving treatment at home should seek medical advice if treatment with SALBUNEB becomes less effective. The dosage or frequency of administration should only be increased on medical advice.

Patients being treated with SALBUNEB may also be receiving other dosage forms of short-acting inhaled bronchodilators to relieve symptoms. Increasing use of bronchodilators, in particular short-acting inhaled Beta₂-agonist to relieve symptoms, indicates deterioration of asthma control. The patient should be instructed to seek medical advice if short-acting relief bronchodilator treatment becomes less effective or more inhalations than usual are required. In this situation patients should be assessed and consideration given to the need for increased anti-inflammatory therapy (e.g. higher doses of inhaled corticosteroid or a course of oral corticosteroid).

Severe exacerbations of asthma must be treated in the normal way.

Salbutamol should be administered cautiously to patients suffering from thyrotoxicosis. Cardiovascular effects may be seen with sympathomimetic drugs, including salbutamol. There is some evidence from post-marketing data and published literature of rare occurrences of myocardial ischaemia associated with salbutamol. Patients with underlying severe heart disease (e.g. ischaemic heart disease, arrhythmia or severe heart failure) who are receiving salbutamol should be warned to seek medical advice if they experience chest pain or other symptoms of worsening heart disease. Attention should be paid to assessment of symptoms such as dyspnoea and chest pain, as they may be of either respiratory or cardiac origin.

SALBUNEB should be used with care in patients known to have received large doses of other sympathomimetic drugs. Potentially serious hypokalaemia may result from Beta₂-agonist therapy, mainly from parenteral and nebulised administration. Particular caution is advised in acute severe asthma as this effect may be potentiated by hypoxia and by concomitant treatment with xanthine derivatives, steroids and diuretics. Serum potassium levels should be monitored in such situations.

In common with other Beta-adrenoceptor agonists, salbutamol can induce reversible metabolic changes such as increased blood glucose levels. Diabetic patients may be unable to compensate for the increase in blood glucose and the development of ketoacidosis has been reported. Concurrent administration of corticosteroids can exaggerate this effect.

Lactic acidosis has been reported in association with high therapeutic doses of intravenous and nebulised short-acting beta-agonist therapy, mainly in patients being treated for an acute asthma exacerbation. Increase in lactate levels may lead to dyspnoea and compensatory hyperventilation, which could be misinterpreted as a sign of asthma treatment failure and lead to inappropriate intensification of short-acting beta-agonist treatment. It is therefore recommended that patients are monitored for the development of elevated serum lactate and consequent metabolic acidosis in this setting.

A small number of cases of acute angle-closure glaucoma have been reported in patients treated with a combination of nebulised Salbutamol and Ipratropium Bromide. A combination of nebulised Salbutamol with nebulised anticholinergics should therefore be used cautiously. Patients should receive adequate instruction in correct administration and be warned not to let the solution or mist enter the eye.

Interactions with Other Medicaments:

Salbutamol and non-selective Beta-blocking drugs such as propranolol should generally not be prescribed together.

Adverse Effects/Undesirable Effects:

Known adverse effect;

Immune system disorders: Hypersensitivity reactions including angioedema, urticaria, bronchospasm, hypotension and collapse.

Metabolism and nutrition disorders: Hypokalaemia, potentially serious hypokalaemia may result from Beta2-agonist, lactic acidosis

Nervous system disorders: Tremor, headache, hyperactivity.

Cardiac disorders: Tachycardia, palpitations, cardiac arrhythmias including atrial fibrillation, supraventricular tachycardia and extrasystoles, myocardial ischaemia

Vascular disorders: Peripheral vasodilatation, respiratory, thoracic and mediastinal disorders, paradoxical bronchospasm.

As with other inhalation therapy, paradoxical bronchospasm may occur with an immediate increase in wheezing after dosing. This should be treated immediately with an alternative presentation or a different fast-acting inhaled bronchodilator.

SALBUNE should be discontinued immediately, the patient assessed, and, if necessary, alternative therapy instituted.

Gastrointestinal disorders: Mouth and throat irritation.

Musculoskeletal and connective tissue disorders: Muscle cramps.

Statement on Usage During Pregnancy and Lactation:

Administration of drugs during pregnancy should only be considered if the expected benefit to the mother is greater than any possible risk to the foetus. As with the majority of drugs, there is little published evidence of the safety of Salbutamol in the early stages of human pregnancy, but in animal studies there was evidence of some harmful effects on the foetus at very high dose levels.

As Salbutamol is probably secreted in breast milk, its use in nursing mothers requires careful consideration. It is not known whether Salbutamol has a harmful effect on the neonate, and so its use should be restricted to situations where it is felt that the expected benefit to the mother is likely to outweigh any potential risk to the neonate.

Overdosage and Treatment:

The most common signs and symptoms of overdose with Salbutamol are transient beta agonist pharmacologically mediated events, including tachycardia, tremor, hyperactivity and metabolic effects including hypokalaemia and lactic acidosis.

Hypokalaemia may occur following overdose with Salbutamol. Serum Potassium levels should be monitored. Consideration should be given to discontinuation of treatment and appropriate symptomatic therapy such as cardioselective beta-blocking agents in patients presenting with cardiac symptoms (e.g. tachycardia, palpitations). Beta-blocking drugs should be used with caution in patients with a history of bronchospasm.

Storage Condition:

Do not store above 30°C. Protect from light.

Shelf life:

2 years from manufacturing date in the proposed storage condition.

Do not use after expiry.

Dosage form and packaging available:

2.5 mL X 60 LDPE plastic ampoule per box.

Manufacturer/Product Registration Holder:

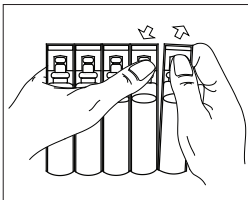


AIN MEDICARE SDN. BHD.
Lot 4933, 4934 & PT2464, Jalan 6/44,
Kaw. Perindustrian Pengkalen Chepa 2,
16100 Kota Bharu, Kelantan, MALAYSIA

HANDLING INSTRUCTION

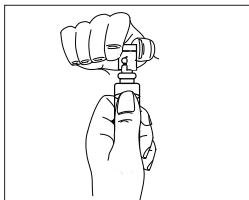
AIN MEDICARE NEBULISER SOLUTION IN POLYETHYLENE (LDPE) AMPOULE

1)



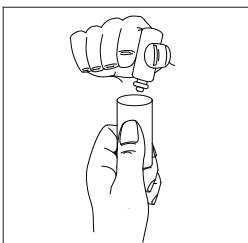
Detach ampoule along parting line

2)



Twist off tab

3)



Squeeze the content of the unit dose vial into the nebuliser reservoir. Assemble the nebuliser and use as directed. After use, throw away any solution left in the reservoir and clean the nebuliser according to the manufacturer's instructions.